

§2.5

An Introduction to Exponents

A shorthand for repeated multiplication, and the rules that come with it

Today's Goal

After completing this section, you should be able to:

- Represent repeated multiplication using exponential notation, identifying the base, the exponent, and the resulting power
- Apply the properties of whole number exponents to simplify numerical expressions written in exponential notation
- Interpret and evaluate a nonzero whole number raised to the zeroth power

Learning Target 7

LT.7 Exponents

I can represent numbers using exponential notation and can perform arithmetic operations that involve exponents.

Shorthand for Repeated Multiplication

In §2.4, multiplication was shorthand for repeated addition.

$$n \times a = \underbrace{a + a + \cdots + a}_{n \text{ addends}}$$

So what should the shorthand for $2 \times 2 \times 2 \times 2 \times 2$ be?

$$2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

Base, Exponent, Power

Definition 2.5.1

If n is a natural number (e.g., 1, 2, 3, etc.) and a is any whole number, then

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}}.$$

In the *exponential notation* a^n , a is the *base* and n is the *exponent*. The number that a^n evaluates to is the n -th *power* of a .

Climbing the Powers of 3

Evaluate 3^2 , 3^3 , 3^4 , and 3^5 .

Power	Previous value $\times 3$	Value
3^2	3×3	9
3^3	9×3	27
3^4	27×3	81
3^5	81×3	243

The 5th power of 3 is 243.

Four Properties of Exponents

Theorem 2.5.3 Properties of Whole Number Exponents

For any whole numbers a and b and any natural numbers m and n (with $m > n$), the following are true.

Product Property: $a^m \times a^n = a^{m+n}$

Quotient Property: $\frac{a^m}{a^n} = a^{m-n}$

Power of a Product Property: $(ab)^m = a^m \times b^m$

Power of a Power Property: $(a^m)^n = a^{m \cdot n}$

Why the Product Property Works

Write out a^m and a^n as factors. How many factors are there in all?

$$\begin{aligned} a^m \times a^n &= \underbrace{a \times a \times \cdots \times a}_{m \text{ factors}} \times \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}} \\ &= \underbrace{a \times a \times \cdots \times a}_{(m+n) \text{ factors}} \\ &= a^{m+n} \end{aligned}$$

The other three come from counting factors the same way.

Adding Exponents, Combining Bases

Simplify. Write your final answers in exponential notation.

a. $4^6 \times 4^{11}$

Same base, so add the exponents: $4^{6+11} = 4^{17}$

b. $3^8 \times 3^5 \times 3^2$

Same base again, so add the exponents: $3^{8+5} \times 3^2 = 3^{13} \times 3^2 = 3^{15}$

c. $3^7 \times 5^7 \times 15^4$

Same exponent, so combine the bases: $(3 \times 5)^7 \times 15^4 = 15^7 \times 15^4 = 15^{11}$

Climbing Back Down

Going up multiplies by 3. Going down divides by 3. So what must 3^0 be?

Power	Value	Step down
3^4	81	
3^3	27	divide by 3
3^2	9	divide by 3
3^1	3	divide by 3
3^0	1	divide by 3

Why Anything to the Zero Is One

Definition 2.5.5

If a is any whole number other than 0, then $a^0 = 1$.

Note the following.

1. In Definition 2.5.5, we must exclude the case where the base is 0 because our observation would require us to divide by 0, which we cannot do.
2. The Product Property in Theorem 2.5.3 still works when an exponent is 0.

Subtracting and Multiplying Exponents

Work through these with your group.

a. $\frac{9^{14}}{9^5}$

Same base, so subtract the exponents: $9^{14-5} = 9^9$

b. $(2^5)^3$

A power of a power, so multiply the exponents: $2^{5 \times 3} = 2^{15}$

c. $\frac{10^{17}}{2^9 \times 5^9}$

Combine the bottom first: $\frac{10^{17}}{10^9} = 10^8$

One Question to Leave With

**Can you simplify $3^5 \times 5^3$
using the properties?**

No. The bases differ, so the Product Property does not apply. The exponents differ, so the Power of a Product Property does not apply either.

A property needs something to match. When nothing matches, all you can do is multiply it out.